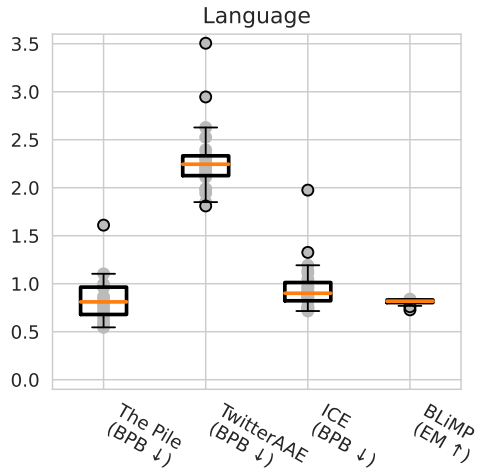

Figure 36: **Targeted evaluation of language. Model accuracy on the four scenarios for evaluating linguistic understanding.**

8.4 Targeted evaluations

Language. To further explore the results for this targeted evaluation, see `https://crfm.stanford.edu/helm/v0.1.0/?group=language` and Figure 36. For the language modeling scenarios, we begin by noting that the models trained on **The Pile** are consistently the most accurate on **The Pile** to no surprise. However, these models also tend to be the most accurate on the other two language modeling scenarios: for **TwitterAAE** and **ICE**, the most accurate models are GPT-J (6B), GPT-NeoX (20B), OPT (66B), OPT (175B), and BLOOM (176B). This suggests that **The Pile**, when compared with the other training sets for these models, may yield better transfer to other language modeling datasets. Further, we are surprised to see the other models trained on **The Pile**, especially Anthropic-LM v4-s3 (52B) and TNLG v2 (530B), which are generally are the most accurate models on the core scenarios (Figure 26), are actually less accurate on **TwitterAAE** and **ICE** compared to models not trained on **The Pile**. Overall, we see that the models that perform most accurately on all language modeling scenarios are quite different and are poorly correlated with the accuracy trends for core scenarios.

Further, since **TwitterAAE** and **ICE** provide unique demographic data, we examine performance disparities for these scenarios. For **TwitterAAE**, we see a clear and consistent trend: all models perform noticeably worse on the African American English subset when compared with the White English subset.[68] Consistent with prior work on other language technologies (Blodgett & OConnor, 2017; Koenecke et al., 2020), this indicates a clear performance disparity between language model performance for African American speakers vs. White speakers along the lines of historical marginalization. All models for the Black subset have BPB above 2.0 (lower is better), whereas almost all models for the White subset are below 1.9, with the best model of OPT (175B) having a BPB of 1.506 for the White English subset and 2.114 for the AAE subset. For **ICE**, across all four regional subsets (East Africa, Hong Kong, India, USA), the most accurate models are the same as those mentioned above for **ICE** and the other language modeling scenarios overall.[69] Further, we see the accuracies for the USA and East Africa tend to be better across all models, with EA being slightly worse, and then India and Hong Kong are clearly worse. For binary gender, we find that the female subset is consistently, but slightly, worse for model accuracy compare to the male subset.

Turning to **BLiMP**, we find that all models achieve similar accuracies. In fact, even on specific subsets for specific linguistic phenomena, we see the accuracies are quite similar. We are surprised to see text-davinci-002 is not one of the most accurate models overall and, instead, is one of the least accurate models on the irregular forms (morphology) and quantifiers (semantics) subsets. Given its consistently high accuracies on

[68]See `https://crfm.stanford.edu/helm/v0.1.0/?group=twitter_aae`.
[69]See `https://crfm.stanford.edu/helm/v0.1.0/?group=ice`.

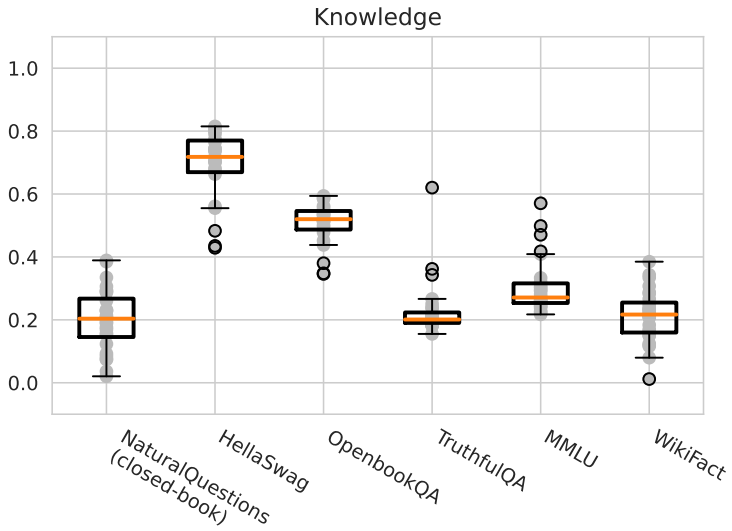

Figure 37: **Targeted evaluation of knowledge.** Model accuracy on the six scenarios (5 question answering, **WikiFact**) for evaluating knowledge acquisition.

various downstream tasks, this may indicate either a con of instruction-tuning or an over-generalization of linguistic rules, especially given the poor performance is on irregular forms in particular.

Knowledge. To further explore the results for this targeted evaluation, see `https://crfm.stanford.edu/helm/v0.1.0/?group=knowledge` and Figure 37. Across all five knowledge-intensive QA scenarios, we see text-davinci-002 is the most accurate. In particular, the gap in accuracies is especially salient for **TruthfulQA**, where text-davinci-002 has an accuracy of 62.0% and the next most accurate model is Anthropic-LM v4-s3 (52B) at 36.2% and for **MMLU**, where text-davinci-002 has an accuracy of 57.0% and the next most acccurate model is Anthropic-LM v4-s3 (52B) at 49.8%. Of these scenarios, for the two that are more centric on factual knowledge (i.e. **MMLU** and **NaturalQuestions** (closed-book)), we see that TNLG v2 (530B) performs notably well, achieving an accuracy within 0.5% of text-davinci-002 for **NaturalQuestions** (closed-book). This aligns clearly with the broader hypothesis that model scale is especially beneficial for memorizing specific factual information, which in turns proves useful for these very knowledge-intensive evaluations.

To further hone in on specific factual knowledge, we then consider model accuracies on **WikiFact**.[70] We again see that text-davinci-002 is the most accurate at 38.5%, with TNLG v2 (530B) as the second most accurate at 34.3%, and with Cohere xlarge v20220609 (52.4B) (33.4%) and davinci (175B) (39.7% as the only other models above 30%. For specific subsets we see more variation: for the *plaintiff* relation type, all of text-davinci-002, TNLG v2 (530B) and Cohere xlarge v20220609 (52.4B) are above 60%, with the next best model being J1-Jumbo v1 (178B) at 51.0% and davinci (175B) being considerably worse (46.5%) despite its overall high accuracy. With that said, in spite of its significantly larger model size, there is no subset where TNLG v2 (530B) performs significantly more accurately than all other models. And all models perform quite poorly for some subsets, e.g. the most accurate model gets an accuracy below 15% for the *discoverer_or_inventor* relation type.

Reasoning. To further explore the results for this targeted evaluation, see `https://crfm.stanford.edu/helm/v0.1.0/?group=reasoning` and Figure 38. Models are most accurate for the structured-data tasks such as entity matching and data imputation (Narayan et al., 2022), which are primarily based on pattern matching and classification. In contrast, models are relatively inaccurate for tasks that involve abstraction, transitive inference, algebraic and logical reasoning, with natural-language tasks such as **LSAT** (Zhong

[70]See `https://crfm.stanford.edu/helm/v0.1.0/?group=wikifact`.

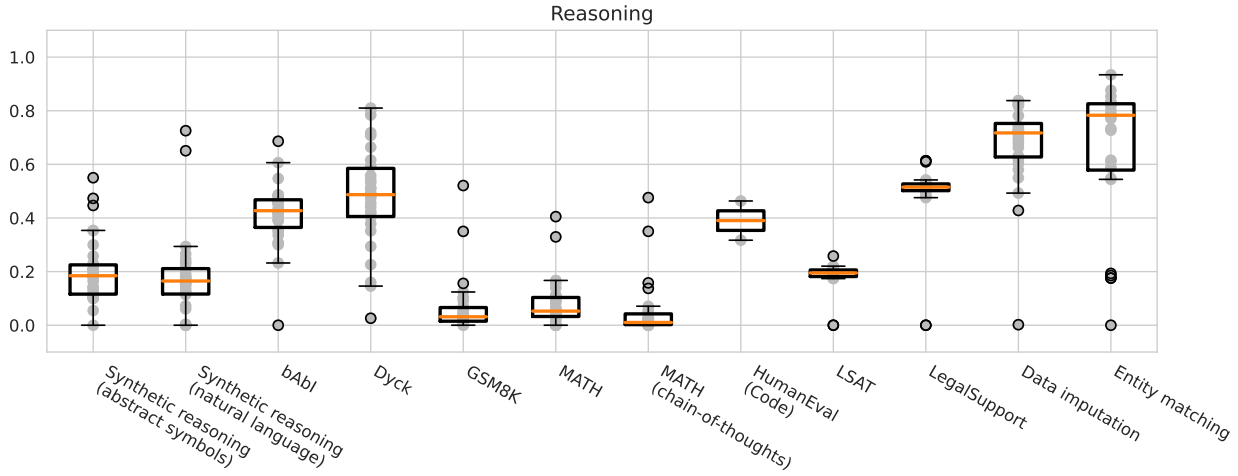

Figure 38: **Targeted evaluation of reasoning.** Model accuracy on 12 scenarios (5 question answering, **WikiFact**) for evaluating reasoning capabilities.

et al., 2021) and **GSM8K** (Cobbe et al., 2020) yielding low accuracies. Overall, we find code-davinci-002 is consistently the most accurate model across reasoning scenarios, even in spite of some scenarios being posed fully in natural language.

For both synthetic reasoning scenarios, we see no model achieves an accuracy above 40% with the exception of text-davinci-002 (47.3% on abstract symbols, 65.1% on natural language) and code-davinci-002 (55.0% on abstract symbols, 67.3% on natural language). That is, these two models show a clear and significant advantage in accuracy for these scenarios, with the gap between them shrinking in the presence of natural language while still maintaining that code-davinci-002 is more accurate than text-davinci-002 for reasoning. We observe that similar trends hold for **MATH**, **GSM8K**, **bAbI**, and **MATH** (chain-of-thoughts). Looking at individual subsets for **bAbI**, we find tasks 3, 4, 15 and 19, which assess transitive reasoning, relational understanding, deduction and planning skills, respectively, to be the the most challenging.[71] In contrast to the trends for text-davinci-002, for Dyck, we observe that text-davinci-002 is not quite accurate (59.4% accuracy), whereas TNLG v2 (530B) (78.4%) joins code-davinci-002 (80.2%) as the only models above 75%.

For **LSAT** (Zhong et al., 2021), which consists of reasoning questions posed for law school admissions, we observe that most evaluated models perform poorly, with accuracies around chance level (20%). Looking into the behavior for individual examples, we see significant variation in behavior that is likely indicative of the spectrum of difficulty of questions. On code scenarios, we see consistent trends with code-davinci-002 consistently outperforming code-cushman-001 (12B) for both **HumanEval** and **APPS**, sometimes by large margins (e.g. 10.% strict correctness vs. 2.6% on **APPS**). We note that we do not evaluate any of text models on these code scenarios, though in some cases this may be sensible/desirable given the striking generality of model development, deployment, and validation/scrutiny. Conversely, while we evaluate the code models for **LSAT** and **LegalSupport**, we find achieve accuracies of 0%. Overall, we find text-davinci-002 and, especially, code-davinci-002 display very strong reasoning capabilities for many different forms of reasoning.

Memorization & Copyright. To further explore the results for this targeted evaluation, see `https://crfm.stanford.edu/helm/v0.1.0/?group=copyright_text`, `https://crfm.stanford.edu/helm/v0.1.0/?group=copyright_code` and Figure 39. We evaluated various models for their ability to reproduce copyrighted text or licensed code. When evaluating source code regurgitation, we only extract from models specialized to code (code-davinci-002 and code-cushman-001 (12B)). When evaluating text regurgitation, we extract from all models except those specialized to code.

[71] See `https://crfm.stanford.edu/helm/v0.1.0/?group=babi_qa`.